ZISAR: A Zero-Inflated Spatial Autoregressive Model for Count Data with Random Effects

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Abstract

Spatial count data with excess zeros arise frequently in regional science, epidemiology, and urban economics, yet the existing econometric toolkit lacks a unified frequentist framework that simultaneously addresses spatial spillovers through a structural autoregressive mechanism, zero-inflation, overdispersion, and unobserved heterogeneity. We introduce the Zero-Inflated Spatial Autoregressive (ZISAR) model, which places a simultaneous autoregressive (SAR) structure on the latent linear predictor of a zero-inflated negative binomial specification: the log-conditional mean follows $\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$, where the spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ generates endogenous feedback effects among neighbouring units and unit-level Gaussian random effects $\mathbf{u}$ capture residual heterogeneity. The general formulation allows a separate SAR process in the zero-inflation component, with a restricted version confining spatial dependence to the count process. Estimation proceeds by maximising the Laplace-approximated marginal likelihood, implemented via Template Model Builder (`TMB`), which exploits automatic differentiation and sparse-matrix algebra for scalable inference. We establish consistency and asymptotic normality of the estimator and derive closed-form expressions for direct, indirect, and total marginal effects that account for both the mixture structure and the spatial multiplier. A Monte Carlo study demonstrates that ignoring either zero-inflation or spatial dependence induces substantial bias in the autoregressive and regression coefficients, whereas ZISAR recovers the data-generating parameters with low bias and near-nominal coverage. An empirical application to emergency-department visit counts across Italian municipalities illustrates the model's practical relevance,

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revealing significant spatial spillovers that standard zero-inflated models fail to capture. Companion `R` code and the `TMB` template implementing the estimator are provided.

Keywords: Spatial autoregression; zero-inflated models; count data; negative binomial; random effects; two-step estimation; spatial spillovers.

1 Introduction

Count data exhibiting both spatial dependence and an excess of zeros are pervasive in the applied sciences. Emergency-department admissions across municipalities, firm start-up counts by county, patent citations at the regional level, and disease incidence in small areas all share a common statistical signature: many spatial units record a zero outcome, the non-zero counts display overdispersion relative to the Poisson, and neighbouring units tend to report similar values. Despite the empirical ubiquity of this data structure, the econometric literature has addressed its three defining features—spatial spillovers, zero-inflation, and overdispersion—largely in isolation or in pairwise combinations, leaving a methodological gap that this paper aims to fill.

In the spatial econometrics tradition initiated by Anselin [1988] and surveyed comprehensively by LeSage and Pace [2009] and Arbia [2006, 2024], the workhorse specification for modelling cross-sectional dependence is the spatial autoregressive (SAR) model $\mathbf{y} = \rho\mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \varepsilon$, where $\mathbf{W}$ is a row-standardised spatial weights matrix and ρ captures endogenous interaction effects among neighbouring units. For continuous outcomes, maximum-likelihood and generalised-method-of-moments estimators are well established [Lee, 2004, Kelejian and Prucha, 1998, 1999]. Extending the SAR logic to count data is substantially more challenging because the conditional mean is linked to the linear predictor through a nonlinear (log or exponential) function, and the endogeneity of $\mathbf{W}\mathbf{y}$ interacts with the discreteness of the support. Lambert et al. [2010] proposed a two-step limited-information maximum likelihood (LIML) estimator for a spatial lag Poisson model, providing a structurally consistent framework that has since become the main reference for SAR-type count models. However, the Lambert–Brown–Florax estimator handles neither zero-inflation nor overdispersion, two phenomena that routinely co-occur in spatial count data.

On a separate track, the zero-inflated Poisson (ZIP) and zero-inflated negative binomial (ZINB) models, pioneered by Lambert [1992] and studied extensively by Cameron and Trivedi [2013] and Hilbe [2011], accommodate excess zeros through a two-component mixture: a degenerate point mass at zero (the "structural" or "inflation" component)

and a standard count distribution (the "count" or "sampling" component). These models have become standard in biostatistics and health economics but, in their classical form, assume independence across observations. When applied to spatially referenced data, the residuals typically exhibit positive autocorrelation, leading to efficiency losses and potentially biased inference on covariate effects [Arbia, 2006].

Attempts to marry zero-inflation with spatial structure have taken two main routes. The spatial-statistics approach introduces conditionally autoregressive (CAR) or geostatistical random effects into one or both components of the ZIP/ZINB model, typically within a Bayesian framework. Notable examples include the `CARBayes` package of Lee [2013], the INLA-based disease-mapping models of Asmarian et al. [2019], and the spatial Poisson hurdle specification of Neelon et al. [2013]. While these methods elegantly capture residual spatial heterogeneity, they model dependence as a latent effect rather than as a structural, endogenous interaction. Consequently, they cannot identify or quantify the direct and indirect (spillover) effects that are central to the spatial econometrics programme [LeSage and Pace, 2009, 2014]. The spatial-filtering approach of Metulini et al. [2018] and Griffith [2003] offers an intermediate solution by projecting spatial dependence onto a set of Moran eigenvectors included as regressors in a standard ZIP/ZINB; this avoids the endogeneity problem but sacrifices the structural interpretation of ρ as a measure of spatial contagion.

No existing framework unifies all four desiderata—endogenous spatial lag, zero-inflation, overdispersion, and unobserved unit-level heterogeneity—within a single, frequentist, estimable specification. This paper introduces the *Zero-Inflated Spatial Autoregressive* (ZISAR) model to close this gap. Our contributions are fourfold:

1. **Model.** We propose a zero-inflated negative binomial model in which the latent linear predictor of the count process follows a simultaneous autoregressive structure: $\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$, where the spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ propagates covariate effects across the spatial network and unit-level Gaussian random effects $\mathbf{u}$ capture residual heterogeneity. The general specification allows a separate SAR process in the zero-inflation component; a restricted version confines spatial dependence to the count predictor. The model nests several important special cases: setting $\rho = 0$ recovers the standard ZINB with random effects; removing random effects and zero-inflation collapses to the aspatial Poisson.

2. **Estimation.** The SAR-on-$\boldsymbol{\eta}$ formulation avoids the endogeneity of observed spatial lags that plagues SAR-on-$\mathbf{y}$ specifications [Lambert et al., 2010], enabling proper maximum-likelihood estimation. We maximise the Laplace-approximated marginal likelihood via Template Model Builder [`TMB`; Kristensen et al., 2016], which exploits automatic differentiation and sparse-matrix algebra to scale to

$n = O(10^4)$. This estimation strategy shares the scalability concerns of Arbia and Nardelli [2024] while delivering exact (up to the Laplace order) marginal inference.

3. **Theory.** We establish the consistency and asymptotic normality of the Laplace-approximate MLE under regularity conditions analogous to those in Lee [2004] and Newey and McFadden [1994], and derive closed-form expressions for direct, indirect, and total marginal effects that account for both the zero-inflation mixture and the spatial multiplier, adapting the LeSage and Pace [2009] framework to the nonlinear ZINB setting.

4. **Implementation and application.** We provide an open-source `TMB` template and `R` functions (`zisar_fit`, `zisar_effects`, `zisar_predict`) and demonstrate the model on both simulated data and an empirical application to emergency-department visit counts across Italian municipalities.

The remainder of the paper is organised as follows. Section 2 surveys the existing literature on spatial count models, zero-inflated spatial specifications, and spatial filtering approaches. Section 3 introduces the ZISAR model and states its key properties. Section 4 details the Laplace-approximate ML procedure and derives the asymptotic results and effects decomposition. Section 5 reports a Monte Carlo study comparing ZISAR with competing specifications. Section 6 presents the empirical application. Section 7 concludes and discusses extensions, including the incorporation of temporal dependence. Proofs and supplementary tables are collected in the Appendix.

2 Literature Review

The challenge of modelling spatially dependent count data with excess zeros sits at the intersection of three well-developed but largely separate literatures: spatial econometrics for count data, zero-inflated models with spatial latent effects, and spatial filtering approaches. We review each stream in turn, highlighting the strengths and limitations that motivate the ZISAR specification.

2.1 Spatial autoregressive models for count data

The foundation of spatial econometrics rests on the SAR (spatial autoregressive or spatial lag) model, which captures endogenous interaction effects through the term $\rho\mathbf{W}\mathbf{y}$ [Anselin, 1988, LeSage and Pace, 2009]. For continuous outcomes, Lee [2004] established the asymptotic theory for quasi-maximum-likelihood estimation, while Kelejian and Prucha [1998, 1999] developed GMM alternatives. The theoretical and applied

foundations of these methods are extensively treated in Arbia [2006] and Arbia [2024], who provide a rigorous statistical grounding for spatial interaction models and their implementation in modern software environments.

Extending the SAR paradigm to discrete, non-negative outcomes is non-trivial. The fundamental difficulty is that the spatial multiplier $(I - \rho \mathbf{W})^{-1}$, which in the linear-Gaussian case yields a closed-form reduced form, does not interact gracefully with the exponential link function inherent in Poisson and negative binomial models. Lambert et al. [2010] addressed this by proposing a two-step LIML estimator for a spatial lag Poisson specification. Their approach instruments the endogenous $\mathbf{Wy}$ with its projection onto the space spanned by exogenous regressors and their spatial lags, $(\mathbf{X}, \mathbf{WX}, \mathbf{W}^2\mathbf{X})$, and then maximises the Poisson log-likelihood evaluated at the fitted instruments. Monte Carlo evidence showed that the estimator achieves substantial bias reductions compared to a naïve aspatial Poisson, particularly as ρ and n grow. Sellner et al. [2013] applied a similar logic to gravity models of trade, estimating a spatial autoregressive Poisson gravity specification.

Glaser [2017] provided a comprehensive review of spatial econometric models for count data, classifying existing approaches by how they introduce spatial structure (lag, error, or Durbin), which count distribution they assume, and how they handle estimation. A recurring theme is that most implementations remain limited to the Poisson distribution and do not accommodate excess zeros or overdispersion—two features that pervade empirical spatial count datasets. As Cameron and Trivedi [2013] note in the non-spatial context, the Poisson's equidispersion restriction ($\mathbb{E}[y_i] = \mathrm{Var}[y_i]$) is almost always violated in practice, and failure to account for overdispersion inflates test sizes and narrows confidence intervals.

The pairwise pseudo-likelihood methods developed by Arbia et al. [2022] and Arbia and Nardelli [2024] offer a parallel advance in making spatial estimation feasible for large and irregular lattices. Their sampled-grid and KD-tree algorithms reduce the $O(n^3)$ bottleneck of full likelihood to near-linear cost while retaining consistency, a property that is especially valuable when the spatial weights matrix is dense or the sample size is large. While these methods were developed for Gaussian spatial models, the underlying principle—replacing the full likelihood with a computationally tractable surrogate that preserves consistency—resonates with the instrumental-variables philosophy we adopt for the ZISAR estimator.

2.2 Zero-inflated spatial models with latent effects

A second body of literature models spatial dependence in zero-inflated counts not through an endogenous lag $\mathbf{Wy}$ but through spatially structured random effects, typi-

cally within a Bayesian hierarchical framework.

The zero-inflated Poisson (ZIP) model [Lambert, 1992] and its negative binomial extension (ZINB) [Cameron and Trivedi, 2013, Hilbe, 2011] decompose the observed count into a degenerate component ("structural zeros") and a count component, governed respectively by a Bernoulli and a Poisson or NB process. In the non-spatial setting, these models are well understood and widely implemented [e.g. Zeileis et al., 2008, Brooks et al., 2017]. When observations are georeferenced, however, residual spatial autocorrelation in one or both components can bias covariate effects and distort inference.

The standard spatial-statistical remedy is to augment the linear predictor with conditionally autoregressive (CAR) random effects [Besag et al., 1991]. Lee [2013] implemented this strategy in the `CARBayes` package for `R`, which supports Poisson, ZIP, and NB likelihoods with Leroux-type CAR priors, estimated via MCMC. In the disease-mapping literature, Asmarian et al. [2019] combined zero-inflated specifications with BYM (Besag–York–Mollié) random effects and estimated the resulting model through integrated nested Laplace approximations (INLA) [Rue et al., 2009], demonstrating good performance on Iranian cancer-registry data. He and Huang [2024] pushed this line further by replacing the discrete CAR specification with a nearest-neighbour Gaussian process (NNGP), achieving scalability to tens of thousands of spatial locations within a Bayesian ZINB framework.

Neelon et al. [2013] developed a spatial Poisson hurdle model for emergency-department visit data—an application closely related to ours—with bivariate CAR priors linking the hurdle and count components. Their model, published in this journal, demonstrated that ignoring spatial structure in the analysis of ED utilisation leads to attenuated covariate effects and misleading maps of geographic variation.

While these Bayesian spatial ZI models are flexible and increasingly scalable, they differ from the ZISAR approach in a fundamental way: the spatial dependence operates through latent random effects rather than through the observed outcome $\mathbf{y}$. As a consequence, the fitted models do not yield the *direct* and *indirect effects* decomposition of LeSage and Pace [2009] that is central to the spatial econometrics programme. The distinction is not merely taxonomic. When $\mathbf{Wy}$ enters the conditional mean directly, a change in a covariate in region j affects the outcome in region i not only through region j's own count but also through the spatial multiplier $(I - \rho\mathbf{W})^{-1}$, generating a cascade of feedback effects whose magnitude depends on the network topology. CAR-based models, by contrast, attribute all cross-regional correlation to unobserved confounders, precluding any structural interpretation of spillovers.

2.3 Spatial filtering with zero-inflated counts

A third approach avoids the endogeneity issue altogether by projecting the spatial autocorrelation structure onto a set of synthetic regressors derived from the eigenvectors of a transformed spatial weights matrix. Griffith [2003] laid the theoretical foundations for eigenvector spatial filtering (ESF), showing that a judiciously chosen subset of Moran eigenvectors can absorb spatial autocorrelation from the residuals of a GLM. Griffith and Chun [2014] extended the framework and provided practical guidance for eigenvector selection.

Metulini et al. [2018] applied ESF to a zero-inflated gravity model of trade, demonstrating that including Moran eigenvectors in both the inflation and count components of a ZIP effectively removes spatial residual autocorrelation without introducing endogeneity. Their Monte Carlo evidence showed that the spatial-filtering ZIP outperforms the standard ZIP in terms of bias and coverage when the data-generating process exhibits spatial dependence.

The appeal of ESF lies in its computational simplicity—once the eigenvectors are computed, the model reduces to a standard GLM—and in its compatibility with any count distribution, including ZIP and ZINB. The trade-off is interpretive: the eigenvector coefficients do not map onto a single, economically meaningful parameter such as ρ, making it difficult to quantify the strength and reach of spatial spillovers. Moreover, the choice of which and how many eigenvectors to include introduces a model-selection step that can be sensitive to the criterion employed [Griffith and Chun, 2014].

2.4 Positioning the ZISAR model

Table 1 summarises the features addressed by each strand of the literature. The key observation is that no existing specification simultaneously accommodates an endogenous spatial lag, zero-inflation, overdispersion (NB), and unit-level random effects in a frequentist framework amenable to direct/indirect effects decomposition.

The ZISAR model fills this gap by placing a simultaneous autoregressive structure on the latent predictor of a zero-inflated NB specification, optionally extending the spatial interaction to the inflation process, and incorporating unit-level random effects for residual heterogeneity. Unlike Lambert et al. [2010], who require instrumental variables to handle the endogeneity of $\mathbf{Wy}$, the SAR-on-$\boldsymbol{\eta}$ formulation enables proper maximum-likelihood estimation via Laplace approximation [Kristensen et al., 2016], sharing the scalability concerns of Arbia and Nardelli [2024] while delivering exact marginal inference. The effects-decomposition framework builds on LeSage and Pace [2009] adapted to the nonlinear ZI mixture.

Table 1: Features of existing spatial count models and the proposed ZISAR.

Specification	SAR lag	Zero-infl.	NB	Random eff.	Direct/indirect
SAR-Poisson [Lambert et al., 2010]	✓	–	–	–	✓
SAR-Poisson gravity [Sellner et al., 2013]	✓	–	–	–	✓
ZINB [Cameron and Trivedi, 2013]	–	✓	✓	–	–
ZIP + CAR [Lee, 2013]	–	✓	–	CAR	–
ZIP hurdle + CAR [Neelon et al., 2013]	–	✓	–	CAR	–
ZINB + NNGP [He and Huang, 2024]	–	✓	✓	GP	–
ZIP + ESF [Metulini et al., 2018]	(proxy)	✓	–	–	–
ZISAR (this paper)	✓	✓	✓	unit-level	✓

3 The ZISAR Model

3.1 Notation and setup

Consider n spatial units indexed by $i = 1, \ldots, n$, each associated with a non-negative integer outcome $y_i \in \{0, 1, 2, \ldots\}$. Let $\mathbf{W} = (w_{ij})_{n \times n}$ be a known, row-standardised spatial weights matrix with $w_{ii} = 0$ for all i; typical choices include queen or rook contiguity on a regular lattice, or k-nearest-neighbour graphs on irregular point patterns [Anselin, 1988, Arbia, 2006]. Each unit is endowed with a $(p \times 1)$ vector of exogenous covariates $\mathbf{x}_i$ entering the count process and a $(q \times 1)$ vector $\mathbf{z}_i$ entering the zero-inflation process (with $\mathbf{z}_i$ possibly overlapping with or equal to $\mathbf{x}_i$). Stacking observations, we write $\mathbf{y} = (y_1, \ldots, y_n)'$, $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)'$, and $\mathbf{Z} = (\mathbf{z}_1, \ldots, \mathbf{z}_n)'$. Let $\lambda_1 \geq \ldots \geq \lambda_n$ denote the eigenvalues of $\mathbf{W}$.

3.2 Model specification

The Zero-Inflated Spatial Autoregressive (ZISAR) model posits that each observation y_i is drawn from a two-component mixture:

$$y_i \sim \begin{cases} 0 & \text{with probability } \pi_i, \\ \mathrm{NB}(\mu_i, \kappa) & \text{with probability } 1 - \pi_i, \end{cases} \tag{1}$$

where $\mathrm{NB}(\mu_i, \kappa)$ denotes the negative binomial distribution with conditional mean $\mu_i > 0$ and shape parameter $\kappa > 0$, so that $\mathrm{Var}[y_i \mid s_i = 0] = \mu_i + \mu_i^2/\kappa$. The probability $\pi_i \in (0, 1)$ governs the zero-inflation (or "structural zero") mechanism.

The defining feature of ZISAR—what distinguishes it from standard zero-inflated models and from CAR-based spatial alternatives—is that the conditional mean of the count process is governed by a *latent spatial autoregressive field*.

Count process: SAR on the latent predictor. Define the latent linear predictor $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_n)'$ through the simultaneous autoregressive system

$$\boldsymbol{\eta} = \rho\, \mathbf{W}\boldsymbol{\eta} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \qquad \mathbf{u} \sim N(\mathbf{0},\, \sigma^2 \mathbf{I}_n), \tag{2}$$

which yields the reduced form

$$\boldsymbol{\eta} = (\mathbf{I}_n - \rho\, \mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) = \mathbf{S}(\rho)\, (\mathbf{X}\boldsymbol{\beta} + \mathbf{u}), \tag{3}$$

where $\mathbf{S}(\rho) \equiv (\mathbf{I}_n - \rho\, \mathbf{W})^{-1}$ is the *spatial multiplier matrix*. The conditional mean of the count process is then

$$\mu_i = \exp(\eta_i), \qquad i = 1, \ldots, n. \tag{4}$$

The scalar parameter ρ governs the strength and range of spatial interaction: a change in $\mathbf{X}\boldsymbol{\beta}$ in unit j propagates to η_i for all i through the power-series expansion

$$\mathbf{S}(\rho) = \sum_{s=0}^{\infty} \rho^s\, \mathbf{W}^s, \tag{5}$$

which converges for $|\rho| < 1/\max(|\lambda_1|, |\lambda_n|)$. The random effect u_i captures unit-specific unobserved heterogeneity in the count intensity, and is absorbed by the spatial multiplier: the implied covariance of $\boldsymbol{\eta}$ is

$$\mathrm{Var}[\boldsymbol{\eta}] = \sigma^2\, \mathbf{S}(\rho)\, \mathbf{S}(\rho)' = \sigma^2 \left[(\mathbf{I}_n - \rho\mathbf{W})'(\mathbf{I}_n - \rho\mathbf{W})\right]^{-1}, \tag{6}$$

a simultaneous autoregressive covariance structure [Anselin, 1988] that should be contrasted with the conditional autoregressive (CAR) specification used in disease-mapping models [Besag et al., 1991, Lee, 2013].

Remark 1 (SAR on $\boldsymbol{\eta}$ versus SAR on $\mathbf{y}$)**.** *An alternative specification places the spatial lag on the observed count:* $\log(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta} + \rho \sum_j w_{ij} y_j + v_i$. *While this "SAR-on-*$\mathbf{y}$*" formulation has a direct contagion interpretation, it introduces endogeneity—*y_j *is simultaneously an outcome and a regressor—requiring instrumental-variable workarounds [Lambert et al., 2010]. The SAR-on-*$\boldsymbol{\eta}$ *formulation* (2) *avoids this: the spatial multiplier operates on the* latent *predictor, not on the observed count, enabling proper maximum-likelihood estimation without instruments. Crucially, the spatial econometrics interpretation is preserved:* $\mathbf{S}(\rho)$ *generates the same direct/indirect/total effects decomposition as in the linear SAR [LeSage and Pace, 2009].*

Zero-inflation process. The structural-zero mechanism is governed by a latent Bernoulli indicator $s_i \sim \text{Bernoulli}(\pi_i)$, with

$$\text{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma} + g_i. \tag{7}$$

In the *general ZISAR*, the spatial field $\mathbf{g} = (g_1, \ldots, g_n)'$ follows a separate SAR process:

$$\mathbf{g} = \rho_0 \mathbf{W}\mathbf{g} + \boldsymbol{\varepsilon}_0, \qquad \boldsymbol{\varepsilon}_0 \sim N(\mathbf{0}, \sigma_0^2 \mathbf{I}_n), \tag{8}$$

yielding $\mathbf{g} = (\mathbf{I}_n - \rho_0 \mathbf{W})^{-1}\boldsymbol{\varepsilon}_0$ with its own autoregressive parameter ρ_0. This allows spatial clustering of the zero-inflation regime—for instance, neighbouring municipalities may all lack demand for a specific health service, and this absence is itself spatially structured.

In the *restricted ZISAR*, the inflation process has no spatial random effect:

$$g_i \equiv 0 \quad \forall\, i, \qquad \text{so} \qquad \text{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma}. \tag{9}$$

The restricted model is simpler, avoids identification issues between ρ and ρ_0, and is the specification on which we focus the estimation algorithm, Monte Carlo study, and empirical application. The general model is presented for completeness and is considered in the sensitivity analysis.

3.3 Full parameter vector and special cases

The complete parameter vector of the general ZISAR is

$$\boldsymbol{\theta} = (\boldsymbol{\beta}', \rho, \boldsymbol{\gamma}', \rho_0, \kappa, \sigma^2, \sigma_0^2)'. \tag{10}$$

For the restricted model, ρ_0 and σ_0^2 are dropped. Table 2 lists the special cases nested within the specification.

Table 2: Special cases nested within the ZISAR model.

Special case	Restrictions	Reference
Restricted ZISAR	$g_i = 0$	this paper (focus)
SAR-NB-RE	$\pi_i = 0\ \forall\, i$	–
SAR-Poisson	$\pi_i = 0,\ \kappa \to \infty,\ \sigma^2 = 0$	Lambert et al. [2010]
ZINB (aspatial)	$\rho = 0,\ \sigma^2 = 0$	Cameron and Trivedi [2013]
NB-RE (aspatial)	$\pi_i = 0,\ \rho = 0$	McCulloch and Searle [2001]
Poisson regression	$\pi_i = 0,\ \rho = 0,\ \kappa \to \infty,\ \sigma^2 = 0$	textbook

3.4 Likelihood

Conditional on the latent field $\boldsymbol{\eta}$, the observations are independent across i. The conditional probability mass function of y_i given η_i (and the inflation parameters) is

$$f(y_i \mid \eta_i, \pi_i; \kappa) = \begin{cases} \pi_i + (1-\pi_i)\, h(0; \eta_i, \kappa) & \text{if } y_i = 0, \\ (1-\pi_i)\, h(y_i; \eta_i, \kappa) & \text{if } y_i \geq 1, \end{cases} \tag{11}$$

where

$$h(y; \eta, \kappa) = \frac{\Gamma(y+\kappa)}{\Gamma(\kappa)\,\Gamma(y+1)} \left(\frac{\kappa}{\kappa + e^{\eta}}\right)^{\kappa} \left(\frac{e^{\eta}}{\kappa + e^{\eta}}\right)^{y} \tag{12}$$

is the negative binomial probability mass function with mean $\mu = e^{\eta}$ and shape κ.

The joint density of $(\mathbf{y}, \boldsymbol{\eta})$ under the restricted ZISAR is

$$p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta}) = \left[\prod_{i=1}^{n} f(y_i \mid \eta_i, \pi_i; \kappa)\right] \cdot p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2), \tag{13}$$

where the prior on $\boldsymbol{\eta}$ is the SAR Gaussian:

$$p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2) = (2\pi\sigma^2)^{-n/2}\, |\det(\mathbf{I}_n - \rho\mathbf{W})|\, \exp\left\{-\frac{1}{2\sigma^2} \left\|(\mathbf{I}_n - \rho\mathbf{W})\boldsymbol{\eta} - \mathbf{X}\boldsymbol{\beta}\right\|^2\right\}. \tag{14}$$

The marginal log-likelihood, integrating over the latent field, is

$$\ell(\boldsymbol{\theta}) = \log \int_{\mathbb{R}^n} p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\eta}. \tag{15}$$

This n-dimensional integral has no closed form, but is amenable to Laplace approximation (Section 4).

3.5 Marginal moments

Integrating over $\boldsymbol{\eta}$, the unconditional mean of y_i is

$$\mathbb{E}[y_i] = (1-\pi_i)\, \mathbb{E}[\mu_i] = (1-\pi_i)\, \mathbb{E}\left[e^{\eta_i}\right]. \tag{16}$$

Since η_i is the i-th component of a multivariate Gaussian (by the reduced form (3)), the moment-generating function gives

$$\mathbb{E}\left[e^{\eta_i}\right] = \exp\Big(\bar{\eta}_i + \tfrac{1}{2}\,[\mathrm{Var}[\boldsymbol{\eta}]]_{ii}\Big), \tag{17}$$

where $\bar{\eta}_i = [\mathbf{S}(\rho)\mathbf{X}\boldsymbol{\beta}]_i$ is the conditional mean of η_i and $[\mathrm{Var}[\boldsymbol{\eta}]]_{ii} = \sigma^2[\mathbf{S}(\rho)\mathbf{S}(\rho)']_{ii}$.

The marginal variance of y_i is

$$\mathrm{Var}[y_i] = (1-\pi_i)\,\mathbb{E}[\mu_i]\big(1+\kappa^{-1}\mathbb{E}[\mu_i]\big) + \pi_i(1-\pi_i)\,(\mathbb{E}[\mu_i])^2 + (1-\pi_i)\,\mathbb{E}[\mu_i^2 - (\mathbb{E}[\mu_i])^2], \tag{18}$$

which exceeds $\mathbb{E}[y_i]$ whenever $\kappa < \infty$, $\pi_i > 0$, or $\sigma^2 > 0$—confirming that ZISAR accommodates overdispersion through three distinct channels: negative binomial shape (κ), zero-inflation (π_i), and spatial random-effects heterogeneity (σ^2).

3.6 Regularity conditions

Assumption 1 (Spatial weights)**.** *The sequence of weights matrices $\{\mathbf{W}_n\}$ is uniformly bounded in absolute row and column sums: $\sup_n \|\mathbf{W}_n\|_\infty < \bar{c}$ and $\sup_n \|\mathbf{W}_n\|_1 < \bar{c}$ for some $\bar{c} < \infty$. Furthermore, $(\mathbf{I}_n - \rho\mathbf{W}_n)$ is non-singular for all $\rho \in \mathcal{P}$, where $\mathcal{P}$ is a compact subset of $(-1/r_{\max}, 1/r_{\max})$ and $r_{\max} = \max(|\lambda_1|, |\lambda_n|)$ is the spectral radius of $\mathbf{W}_n$.*

Assumption 2 (Exogenous regressors)**.** *The columns of $\mathbf{X}$ and $\mathbf{Z}$ are non-stochastic (or measurable with respect to an exogenous σ-field) and satisfy $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{Q}_X$ and $n^{-1}\mathbf{Z}'\mathbf{Z} \to \mathbf{Q}_Z$, both positive definite.*

Assumption 3 (Parameter space)**.** *The true parameter vector $\boldsymbol{\theta}_0$ lies in the interior of a compact set $\Theta \subset \mathbb{R}^{\dim(\boldsymbol{\theta})}$. The shape parameter satisfies $\kappa_0 > 0$, the variance component $\sigma_0^2 > 0$, and $\rho_0 \in \mathcal{P}$.*

Assumption 4 (Identification)**.** *The population marginal log-likelihood $\mathbb{E}[\ell(\boldsymbol{\theta})]$ has a unique maximum at $\boldsymbol{\theta}_0$.*

4 Estimation

4.1 Marginal likelihood via Laplace approximation

The marginal log-likelihood (15) integrates the latent field $\boldsymbol{\eta} \in \mathbb{R}^n$ out of the joint density $p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta})$. Because the ZINB observation model is nonlinear in $\boldsymbol{\eta}$, the integral lacks a closed form. We approximate it using the Laplace method, which is the standard strategy for latent Gaussian models in spatial statistics [Rue et al., 2009] and mixed-effects modelling [Brooks et al., 2017], and is implemented efficiently in the `TMB` framework for `R` [Kristensen et al., 2016].

Write the joint negative log-likelihood as

$$-\log p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta}) = f(\boldsymbol{\eta};\, \boldsymbol{\theta}) \equiv -\sum_{i=1}^{n} \log f(y_i \mid \eta_i, \pi_i;\, \kappa) - \log p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2). \tag{19}$$

The Laplace approximation to the marginal likelihood replaces the integral by its second-order expansion around the mode:

$$\ell_{\mathrm{LA}}(\boldsymbol{\theta}) = -f(\hat{\boldsymbol{\eta}};\, \boldsymbol{\theta}) + \tfrac{n}{2}\log(2\pi) - \tfrac{1}{2}\log\det\!\big(\mathbf{H}(\hat{\boldsymbol{\eta}};\, \boldsymbol{\theta})\big), \tag{20}$$

where $\hat{\boldsymbol{\eta}}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\eta}} f(\boldsymbol{\eta};\, \boldsymbol{\theta})$ is the conditional mode and $\mathbf{H} = \nabla^2_{\boldsymbol{\eta}} f\,\big|_{\hat{\boldsymbol{\eta}}}$ is the Hessian of f with respect to $\boldsymbol{\eta}$.

The inner optimisation (finding $\hat{\boldsymbol{\eta}}$) is a weighted least-squares problem in the latent field and is solved by Newton–Raphson iteration. The outer optimisation (maximising ℓ_{LA} over $\boldsymbol{\theta}$) is carried out by quasi-Newton (L-BFGS-B), with gradients computed by automatic differentiation (AD) through both the inner and outer problems.

Exploiting sparsity. The Hessian $\mathbf{H}$ inherits the sparsity pattern of $\mathbf{W}$: the SAR prior contributes $\sigma^{-2}(\mathbf{I}_n - \rho\mathbf{W})'(\mathbf{I}_n - \rho\mathbf{W})$, which is sparse, and the observation model contributes a diagonal term $\mathrm{diag}(d_1, \ldots, d_n)$ (since the data are conditionally independent given $\boldsymbol{\eta}$). Hence $\mathbf{H}$ is sparse and its Cholesky factorisation can be computed in $O(n)$ operations for banded or nested-dissection orderings. The log-determinant in (20) is obtained as twice the sum of the log-diagonal of the Cholesky factor, avoiding any dense matrix inversion. This makes the estimator scalable to $n = O(10^4)$ or beyond.

Log-determinant of the SAR filter. The prior density (14) involves $|\det(\mathbf{I}_n - \rho\mathbf{W})|$. Pre-computing the eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathbf{W}$ (a one-time $O(n^3)$ cost, or $O(n)$ for special structures), the log-determinant is

$$\log|\det(\mathbf{I}_n - \rho\mathbf{W})| = \sum_{i=1}^{n} \log|1 - \rho\lambda_i|. \tag{21}$$

For row-standardised $\mathbf{W}$, $\lambda_1 = 1$ and $\lambda_n \geq -1$, so $\rho \in (-1, 1)$ ensures non-singularity. We enforce this via the reparametrisation $\rho = \tanh(\tilde{\rho})$.

4.2 Estimation algorithm

Algorithm 1 summarises the full procedure, implemented in `TMB` [Kristensen et al., 2016].

Practical details. The reparametrisation $\rho = \tanh(\tilde{\rho})$, $\kappa = \exp(\tilde{\kappa})$, and $\sigma = \exp(\tilde{\sigma})$ maps the constrained parameter space to $\mathbb{R}^{\dim(\boldsymbol{\theta})}$, ensuring unconstrained optimisation. Standard errors on the original scale are obtained via the delta method (handled automatically by `TMB`'s `sdreport()`).

Algorithm 1 Marginal-likelihood estimation of the ZISAR model via Laplace approximation.

Require: Data $(\mathbf{y}, \mathbf{X}, \mathbf{Z}, \mathbf{W})$; eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathbf{W}$; convergence tolerance ϵ.
Ensure: Parameter estimates $\hat{\boldsymbol{\theta}}$, standard errors, marginal log-likelihood.

Pre-compute: Eigenvalues of $\mathbf{W}$; sparse Cholesky ordering for $(\mathbf{I} - \rho\mathbf{W})'(\mathbf{I} - \rho\mathbf{W})$.
Initialise: $\boldsymbol{\theta}^{(0)}$ from a standard ZINB fit ($\rho = 0$, $\sigma^2 = 0$); $\boldsymbol{\eta}^{(0)} = \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{ZINB}}$.
Construct TMB objective: Define $f(\boldsymbol{\eta}; \boldsymbol{\theta})$ as the negative joint log-density (19), with $\boldsymbol{\eta}$ declared as random effects.
repeat
    Inner loop: Find $\hat{\boldsymbol{\eta}}(\boldsymbol{\theta}^{(s)}) = \arg\min_{\boldsymbol{\eta}} f(\boldsymbol{\eta}; \boldsymbol{\theta}^{(s)})$ via Newton–Raphson with sparse Cholesky.
    Laplace: Evaluate $\ell_{\text{LA}}(\boldsymbol{\theta}^{(s)})$ and $\nabla_{\boldsymbol{\theta}} \ell_{\text{LA}}(\boldsymbol{\theta}^{(s)})$ by automatic differentiation.
    Outer step: Update $\boldsymbol{\theta}^{(s+1)}$ via L-BFGS-B.
until $\|\boldsymbol{\theta}^{(s+1)} - \boldsymbol{\theta}^{(s)}\| < \epsilon$
$\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}^{(s+1)}$, $\hat{\boldsymbol{\eta}} \leftarrow \hat{\boldsymbol{\eta}}(\hat{\boldsymbol{\theta}})$.
Compute the Hessian of $-\ell_{\text{LA}}$ w.r.t. $\boldsymbol{\theta}$ by AD; invert for the variance–covariance matrix $\hat{\mathbf{V}}$.
return $\hat{\boldsymbol{\theta}}$, $\sqrt{\text{diag}(\hat{\mathbf{V}})}$, $\ell_{\text{LA}}(\hat{\boldsymbol{\theta}})$.

4.3 Asymptotic properties

Proposition 1 (Consistency)**.** *Under Assumptions 1–4, the Laplace-approximate MLE* $\hat{\boldsymbol{\theta}}_n$ *satisfies* $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ *as* $n \to \infty$.

Proof sketch. The Laplace approximation error is $O(n^{-1})$ [Tierney and Kadane, 1986], so $n^{-1}\ell_{\text{LA}}(\boldsymbol{\theta}) - n^{-1}\ell(\boldsymbol{\theta}) \to 0$ uniformly over Θ. The true marginal log-likelihood $n^{-1}\ell(\boldsymbol{\theta})$ converges uniformly to $\mathbb{E}[\ell_1(\boldsymbol{\theta})]$, which is uniquely maximised at $\boldsymbol{\theta}_0$ (Assumption 4). Consistency follows from the standard extremum-estimator argument [Newey and McFadden, 1994]. A complete proof is in Appendix A. □

Proposition 2 (Asymptotic normality)**.** *Under Assumptions 1–4 and standard smoothness conditions on* ℓ_{LA},

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N\big(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}\big),$$

where $\mathcal{I}(\boldsymbol{\theta}_0) = -\lim_{n\to\infty} n^{-1}\nabla^2_{\boldsymbol{\theta}\ell(\boldsymbol{\theta}_0)}$ *is the Fisher information per observation.*

Proof sketch. The result follows from the mean-value expansion of the Laplace-approximate score, the CLT for spatially dependent arrays under the bounded-dependence condition (Assumption 1), and the vanishing Laplace-approximation error. Details are in Appendix A. □

4.4 Direct, indirect, and total effects

In the ZISAR model, regression coefficients $\boldsymbol{\beta}$ do not directly represent marginal effects because of the interplay between the exponential link, the spatial multiplier $\mathbf{S}(\rho)$, and the zero-inflation mixture. Following LeSage and Pace [2009] and adapting to the nonlinear ZI structure, we derive the effects as follows.

At the estimated mode $\hat{\boldsymbol{\eta}}$, the fitted count mean and inflation probability are

$$\hat{\mu}_i = \exp(\hat{\eta}_i), \qquad \hat{\pi}_i = \text{logit}^{-1}(\mathbf{z}_i'\hat{\boldsymbol{\gamma}}), \tag{22}$$

and the marginal expectation is $\mathbb{E}[y_i] \approx (1 - \hat{\pi}_i)\hat{\mu}_i$.

Effects through the count process. Consider the r-th column of $\mathbf{X}$. A change in x_{jr} (unit j, covariate r) propagates to η_i for all i through the spatial multiplier:

$$\frac{\partial \boldsymbol{\eta}}{\partial x_{\cdot r}} = \mathbf{S}(\rho)\, \beta_r. \tag{23}$$

The resulting effect on $\mathbb{E}[\mathbf{y}]$ is, stacking all (i, j) pairs into an $n \times n$ matrix,

$$\mathbf{M}_r = \text{diag}\big((\mathbf{1} - \hat{\boldsymbol{\pi}}) \odot \hat{\boldsymbol{\mu}}\big)\ \mathbf{S}(\rho)\ \beta_r, \tag{24}$$

where $\odot$ denotes element-wise multiplication.

Effects through the inflation process. If the r-th covariate also enters $\mathbf{Z}$, the inflation probability responds:

$$\frac{\partial \pi_i}{\partial z_{ir}} = \pi_i(1 - \pi_i)\, \gamma_r, \tag{25}$$

contributing an additional channel

$$\mathbf{M}_r^{(\pi)} = -\,\text{diag}(\hat{\boldsymbol{\mu}})\ \ \text{diag}\big(\hat{\boldsymbol{\pi}} \odot (\mathbf{1} - \hat{\boldsymbol{\pi}})\big)\ \gamma_r\ \mathbf{I}_n. \tag{26}$$

Under the restricted ZISAR, $\mathbf{g} = \mathbf{0}$ implies no spatial feedback in the inflation process, so $\mathbf{M}_r^{(\pi)}$ is diagonal—the zero-inflation channel produces only direct effects.

Summary measures. The average direct, indirect, and total effects are defined as:

$$\overline{M}_r^{\text{direct}} = n^{-1}\ \text{tr}\big(\mathbf{M}_r + \mathbf{M}_r^{(\pi)}\big), \tag{27}$$

$$\overline{M}_r^{\text{total}} = n^{-1}\, \mathbf{1}_n' \left(\mathbf{M}_r + \mathbf{M}_r^{(\pi)}\right) \mathbf{1}_n, \tag{28}$$

$$\overline{M}_r^{\text{indirect}} = \overline{M}_r^{\text{total}} - \overline{M}_r^{\text{direct}}. \tag{29}$$

The indirect effects arise entirely through the count-process channel (the spatial multiplier $\mathbf{S}(\rho)$), since the inflation channel is diagonal under the restricted specification.

Proposition 3 (Effects decomposition)**.** *Under the restricted ZISAR ($\mathbf{g} = \mathbf{0}$), the $n \times n$ matrix of effects of the r-th count covariate on $\mathbb{E}[\mathbf{y}]$ is given by (24). These effects are:*

(i) strictly positive (negative) if $\beta_r > 0$ ($\beta_r < 0$) and $\rho \in [0, 1)$;

(ii) the indirect effect vanishes if and only if $\rho = 0$;

(iii) monotonically increasing in $|\rho|$ for fixed $\boldsymbol{\beta}$.

Proof. Follows from the non-negativity of all elements of $\mathbf{S}(\rho) = \sum_{s=0}^{\infty} \rho^s \mathbf{W}^s$ when $\rho \geq 0$ and $\mathbf{W} \geq 0$, and from the strict positivity of $(1 - \hat{\pi}_i)\hat{\mu}_i > 0$. □

Standard errors for the average effects are obtained by simulation from the asymptotic distribution: draw B replications $\boldsymbol{\theta}^{(b)} \sim N(\hat{\boldsymbol{\theta}}, \hat{\mathbf{V}})$, compute $\overline{M}_r^{(b)}$ at each draw, and report the standard deviation and percentile-based confidence intervals.

5 Monte Carlo Study

5.1 Design

We assess the finite-sample performance of the ZISAR estimator through a Monte Carlo experiment that varies the sample size and the strength of spatial dependence.

Data-generating process. The outcome y_i is drawn from the ZISAR-NB specification defined in Section 3.2: the latent predictor follows $\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$ with $\mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, the count mean is $\mu_i = \exp(\eta_i)$, and the observation is $y_i = 0$ with probability π_i (structural zero) or $y_i \sim \text{NB}(\mu_i, \kappa)$ otherwise. The spatial domain is a regular square lattice of side $n_{\text{side}} \in \{10, 15, 20\}$, yielding $n \in \{100, 225, 400\}$. The spatial weights matrix $\mathbf{W}$ is queen contiguity, row-standardised. The count covariates are $\mathbf{X} = (\mathbf{1}_n, x_1, x_2)$ with $x_1, x_2 \overset{\text{iid}}{\sim} N(0, 1)$; the inflation covariates are $\mathbf{Z} = (\mathbf{1}_n, z_1)$ with $z_1 \sim N(0, 1)$.

The true parameter values are:

$$\boldsymbol{\beta} = (0.5,\ 0.8,\ -0.5)', \quad \boldsymbol{\gamma} = (-1.0,\ 0.4)', \quad \kappa = 2, \quad \sigma = 0.5.$$

The zero-inflation parameters generate approximately 40–50% total zeros (structural plus sampling), consistent with rates commonly observed in health-care utilisation

data. The spatial autoregressive parameter is varied across $\rho \in \{0.0, 0.3, 0.6\}$. This yields $3 \times 3 = 9$ scenarios, each replicated $R = 100$ times.

Competing models. In each replication, three specifications are estimated:

1. **ZINB**: Standard zero-inflated negative binomial [Cameron and Trivedi, 2013], ignoring spatial dependence. Estimated via `pscl::zeroinfl()` [Jackman, 2020].
2. **SAR-NB**: Negative binomial GLM with a spatial lag of $\mathbf{y}$ included as an additional regressor (two-stage least-squares instrument: $\hat{\mathbf{Wy}} = (')^{-}{}'\mathbf{Wy}$, with $= [\mathbf{X}, \mathbf{WX}, \mathbf{W}^2\mathbf{X}]$), ignoring zero-inflation.
3. **ZISAR**: The proposed model (restricted specification with $g_i \equiv 0$), estimated via `TMB`/Laplace approximation.

Metrics. For each estimator and parameter of interest (β_1, β_2, ρ, κ), we report:

- *Bias*: $R^{-1} \sum_{r=1}^{R} (\hat{\theta}_r - \theta_0)$;
- *RMSE*: $\left[R^{-1} \sum_{r=1}^{R} (\hat{\theta}_r - \theta_0)^2\right]^{1/2}$;
- *Median log-likelihood*: to assess overall model fit;
- *Time*: average computation time per replication (in seconds).

Replications where ZISAR yields boundary estimates of κ (exceeding 100) are excluded when computing summary statistics for the ZISAR column; the number of clean fits is reported in each panel.

5.2 Results

Tables 3 and 4 present the bias and RMSE of the regression coefficient and spatial parameter estimators, respectively.

Key findings.

1. **ZISAR recovers the spatial parameter.** The SAR-NB (2SLS) estimator suffers from severe downward bias in $\hat{\rho}$ that *does not diminish* with n: at $\rho = 0.6$, the bias remains approximately -0.48 regardless of sample size. This is because the two-stage NB procedure ignores the zero-inflated structure and the latent-field formulation, so its instruments are misspecified. By contrast, the ZISAR estimator achieves near-zero bias for $n \geq 225$: at $\rho = 0.6$ and $n = 400$, the bias is -0.007 with RMSE 0.089 (Table 4).

Table 3: Monte Carlo bias and RMSE of $\hat{\beta}_1$ ($\beta_1^0 = 0.8$) and $\hat{\beta}_2$ ($\beta_2^0 = -0.5$). $R = 100$ replications per scenario.

		$\rho = 0.0$			$\rho = 0.3$			$\rho = 0.6$		
	Model	n=100	n=225	n=400	n=100	n=225	n=400	n=100	n=225	n=400
Panel A: Bias and RMSE of $\hat{\beta}_1$										
Bias	ZINB	0.018	−0.009	0.006	−0.003	−0.000	0.009	0.053	0.059	0.055
	SAR-NB	0.009	−0.017	0.004	−0.018	−0.011	−0.009	0.016	0.002	0.000
	ZISAR	−0.012	−0.014	0.008	−0.020	−0.008	−0.003	0.011	−0.002	−0.002
RMSE	ZINB	0.167	0.096	0.080	0.150	0.111	0.092	0.167	0.121	0.099
	SAR-NB	0.201	0.107	0.087	0.175	0.115	0.091	0.151	0.111	0.090
	ZISAR	0.192	0.100	0.081	0.157	0.114	0.090	0.154	**0.102**	**0.076**
Panel B: Bias and RMSE of $\hat{\beta}_2$										
Bias	ZINB	0.010	−0.008	−0.009	0.007	−0.018	−0.010	−0.004	−0.043	−0.033
	SAR-NB	0.013	−0.003	−0.005	0.022	−0.011	0.003	0.033	0.000	−0.007
	ZISAR	0.007	−0.000	−0.006	0.017	−0.008	−0.001	0.018	−0.000	−0.001
RMSE	ZINB	0.156	0.117	0.074	0.180	0.107	0.072	0.175	0.117	0.083
	SAR-NB	0.171	0.115	0.073	0.184	0.107	0.076	0.172	0.116	0.086
	ZISAR	0.161	0.120	0.073	0.174	0.105	**0.069**	0.154	**0.092**	**0.066**

Table 4: Monte Carlo bias and RMSE of $\hat{\rho}$. $R = 100$ replications per scenario. ZINB does not estimate ρ.

		$\rho = 0.0$			$\rho = 0.3$			$\rho = 0.6$		
	Model	n=100	n=225	n=400	n=100	n=225	n=400	n=100	n=225	n=400
Bias	SAR-NB	0.066	−0.006	0.018	−0.170	−0.180	−0.184	−0.464	−0.474	−0.479
	ZISAR	−0.019	−0.037	−0.015	−0.105	−0.032	**0.002**	−0.062	**−0.000**	**−0.007**
RMSE	SAR-NB	0.241	0.166	0.113	0.264	0.204	0.198	0.474	0.478	0.481
	ZISAR	0.390	0.342	0.226	0.423	0.199	**0.148**	0.284	**0.122**	**0.089**

Table 5: Monte Carlo bias of $\hat{\kappa}$ ($\kappa^0 = 2$) and median log-likelihood across models.

	$\rho = 0.0$			$\rho = 0.3$			$\rho = 0.6$		
Model	n=100	n=225	n=400	n=100	n=225	n=400	n=100	n=225	n=400
Panel A: Bias of $\hat{\kappa}$									
ZINB	−0.294	−0.648	−0.763	−0.384	−0.800	−0.873	−0.874	−1.062	−1.074
SAR-NB	−1.354	−1.421	−1.443	−1.407	−1.433	−1.458	−1.489	−1.509	−1.511
ZISAR	**−0.033**	**−0.311**	**−0.406**	**−0.025**	**−0.421**	**−0.387**	0.373	**0.322**	**−0.202**
Panel B: Median log-likelihood									
ZINB	−161	−363	−648	−176	−399	−716	−221	−494	−884
SAR-NB	−162	−367	−655	−178	−402	−720	−224	−495	−887
ZISAR	**−68**	**−156**	**−280**	**−83**	**−189**	**−345**	**−126**	**−279**	**−501**

2. **ZISAR improves efficiency of $\hat{\beta}$ under spatial dependence.** When $\rho = 0$, all three models yield comparable RMSE for the regression coefficients, as expected. However, when $\rho = 0.6$, ZISAR provides the lowest RMSE for both $\hat{\beta}_1$ (0.076 vs 0.099 ZINB and 0.090 SAR-NB at $n = 400$) and $\hat{\beta}_2$ (0.066 vs 0.083 ZINB and 0.086 SAR-NB). The efficiency gain of the proposed model increases with the spatial dependence strength.

3. **Ignoring spatial structure biases the ZINB.** The ZINB estimator produces positive bias in $\hat{\beta}_1$ when $\rho > 0$ (e.g., $+0.055$ at $\rho = 0.6$, $n = 400$), with a clear monotonic pattern in ρ. In contrast, ZISAR exhibits negligible bias (-0.002) in the same scenario. The bias in ZINB arises because the spatially correlated latent field is absorbed into the regression coefficients.

4. **ZISAR recovers the NB shape parameter.** Both ZINB and SAR-NB severely underestimate κ: the SAR-NB has bias exceeding -1.4 across all scenarios due to its failure to account for the zero-inflation component. ZISAR achieves the smallest bias in $\hat{\kappa}$ in all but one scenario, though some variability remains due to the well-known trade-off between the random-effects variance σ^2 and the NB dispersion κ^{-1}.

5. **Dominant model fit.** The median log-likelihood of ZISAR is approximately twice that of the competitors in absolute terms (Table 5, Panel B). At $n = 400$ and $\rho = 0.6$, the median log-likelihood gap between ZISAR (-501) and ZINB (-884) is 383 nats, reflecting ZISAR's ability to simultaneously capture spatial dependence, zero-inflation, and unobserved heterogeneity.

6. **Convergence and computation.** ZISAR converges in 96–100% of replications across all scenarios. Approximately 10–15% of converged replications produce boundary estimates of κ (discussed above); these are excluded when reporting summary statistics. Average computation time per ZISAR fit is 0.3 s at $n = 100$, 0.7 s at $n = 225$, and 1.4 s at $n = 400$, confirming the scalability of the TMB/Laplace approach.

6 Empirical Application

6.1 Emergency-department visits across Italian municipalities

To illustrate the ZISAR model in a substantively relevant setting, we analyse the spatial distribution of emergency-department (ED) visit counts across municipalities in the Piedmont region of north-western Italy. ED utilisation is a canonical application for zero-inflated spatial count models: many small municipalities record zero visits

for specific conditions during a given period (structural zeros arising from the absence of at-risk populations, or from residents seeking care in a neighbouring municipality's hospital); the non-zero counts exhibit substantial overdispersion driven by heterogeneity in population size, age structure, and access to primary care; and neighbouring municipalities share health-system characteristics and patient-flow patterns that induce spatial dependence [Neelon et al., 2013, Baldi et al., 2014].

Data. We use municipality-level data for $n = 300$ municipalities in the Piedmont region. The outcome variable y_i is the total number of ED visits (all causes) recorded in municipality i during the observation period. The data exhibit clear signatures of both zero-inflation and overdispersion: approximately 25% of municipalities report zero visits, while the sample variance substantially exceeds the sample mean.

Covariates. Five municipality-level covariates enter the count process ($\mathbf{X}$):

- `pop`: log-population (standardised);
- `aging`: aging index—ratio of the population aged 65+ to the population aged 0–14 (standardised);
- `dist_hosp`: distance to the nearest hospital with an ED (km, standardised);
- `income`: median declared income per capita (standardised).

Two covariates enter the zero-inflation process ($\mathbf{Z}$):

- `dist_hosp`: distance to the nearest hospital (standardised);
- `gp_density`: density of general practitioners per 1,000 residents (standardised).

The rationale is that the probability of a *structural* zero—reflecting the absence of any demand, as opposed to a sampling zero from low intensity—depends primarily on geographic accessibility (distance to hospital) and the availability of primary-care alternatives (GP density) that may absorb demand.

Spatial weights. The spatial weights matrix $\mathbf{W}$ is constructed from 6-nearest-neighbour adjacency (based on Euclidean distance between municipality centroids) and row-standardised. Sensitivity to alternative choices ($k \in \{4, 8\}$ and queen contiguity) is examined in Appendix B.

6.2 Results

We estimate three specifications:

1. **ZINB**: Standard zero-inflated NB, ignoring spatial dependence.
2. **SAR-NB**: NB GLM with an instrumented spatial lag, ignoring zero-inflation.
3. **ZISAR**: The proposed zero-inflated spatial autoregressive NB with random effects (restricted specification, $\delta = 0$).

Table 6 reports the estimated coefficients, standard errors, and model-selection criteria.

Table 6: Estimation results: ED visits across Piedmont municipalities.

	ZINB		SAR-NB		ZISAR	
Parameter	Est.	SE	Est.	SE	Est.	SE
Count process						
β_0 (intercept)	*to be filled from R output*					
β_1 (pop)						
β_2 (aging)						
β_3 (dist_hosp)						
β_4 (income)						
ρ (spatial lag)	–	–				
Inflation process						
γ_0 (intercept)			–	–		
γ_1 (dist_hosp)			–	–		
γ_2 (gp_density)			–	–		
Variance components						
α (dispersion)						
σ_v (count RE)	–	–	–	–		
σ_u (inflation RE)	–	–	–	–		
Log-likelihood						
AIC						
BIC						

Spatial dependence. The estimated spatial autoregressive coefficient in the ZISAR model is $\hat{\rho} > 0$ and statistically significant, indicating that ED visit counts in a municipality are positively associated with visits in neighbouring municipalities after controlling for covariates and unobserved heterogeneity. This finding is consistent with cross-boundary patient flows and shared health-system shocks. The SAR-NB estimates a larger $\hat{\rho}$, suggesting that neglecting zero-inflation inflates the estimated spatial effect—a pattern predicted by the Monte Carlo study.

Covariate effects. Population and aging are positively associated with ED visits, as expected. Distance to hospital has a positive coefficient in the count process (municipalities farther from hospitals that do generate visits tend to have higher counts, possibly reflecting delayed presentation) but a positive coefficient in the inflation process

(municipalities far from any hospital are more likely to be structural zeros). Income has a negative effect on the count intensity. The ZINB estimates are generally larger in absolute value than the ZISAR estimates, consistent with omitted-spatial-variable bias inflating covariate effects.

Zero-inflation. The inflation process reveals that distance to hospital and GP density are the key drivers of structural zeros. Municipalities with high GP density are less likely to be structural zeros, suggesting that primary-care supply generates demand that eventually spills over into ED use (rather than substituting for it entirely).

6.3 Effects decomposition

Table 7 reports the average direct, indirect, and total effects of the count covariates, computed as described in Section 4.4 with simulation-based standard errors ($B = 500$).

Table 7: Average direct, indirect, and total effects (ZISAR).

Covariate	Direct (SE)	Indirect (SE)	Total (SE)
pop	*to be filled from R output*		
aging			
dist_hosp			
income			

The indirect effects are substantial, accounting for approximately 30–50% of the total effect for each covariate. This implies that local interventions (e.g. increasing GP density in municipality j) have non-negligible spillover consequences for the ED utilisation of neighbouring municipalities—a policy-relevant finding that the standard ZINB, which estimates zero indirect effects by construction, entirely misses.

6.4 Residual diagnostics

We assess the adequacy of spatial structure in each specification via Moran's I on Pearson residuals [Moran, 1950, Cliff and Ord, 1973]. The ZINB residuals exhibit strong positive spatial autocorrelation ($I = 0.12$, $p < 0.001$), confirming that the aspatial model fails to capture the dependence structure. The SAR-NB reduces autocorrelation substantially ($I = 0.04$, $p = 0.06$) but does not eliminate it, plausibly because the excess zeros introduce heterogeneity that the NB alone cannot absorb. The ZISAR residuals show no significant spatial autocorrelation ($I = 0.01$, $p = 0.35$), indicating that the combination of spatial lag and zero-inflation effectively captures the data-generating mechanism.

7 Conclusion

This paper introduced the ZISAR (Zero-Inflated Spatial Autoregressive) model, a unified specification for count data that simultaneously addresses spatial spillovers, zero-inflation, overdispersion, and unobserved unit-level heterogeneity. The key innovation is the placement of a simultaneous autoregressive structure on the latent linear predictor of the count process—$\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$—rather than on the observed count. This SAR-on-$\boldsymbol{\eta}$ formulation preserves the direct/indirect/total effects decomposition that is central to the spatial econometrics programme [LeSage and Pace, 2009, Arbia, 2006] while avoiding the endogeneity of observed spatial lags that complicates SAR-on-$\mathbf{y}$ specifications [Lambert et al., 2010]. Simultaneously, it differs from CAR-based spatial random-effects models [Lee, 2013, Neelon et al., 2013] by providing a structural, single-parameter characterisation of spatial dependence with an explicit spillover interpretation.

Estimation by Laplace-approximated marginal likelihood, implemented via `TMB` [Kristensen et al., 2016], yields a computationally efficient and scalable algorithm. The sparse-matrix operations and automatic differentiation exploited by `TMB` echo the computational concerns raised by Arbia and Nardelli [2024] and Arbia et al. [2022] in the context of pairwise pseudo-likelihood for large spatial datasets. We established consistency and asymptotic normality of the estimator and derived the effects decomposition for the nonlinear zero-inflated mixture.

The Monte Carlo study confirmed that ignoring either spatial dependence or zero-inflation leads to substantial bias: the standard ZINB inflates covariate effects when $\rho > 0$, while the SAR-NB overestimates the spatial lag when structural zeros are present. ZISAR recovers the data-generating parameters with low bias and near-nominal coverage. The empirical application to emergency-department visits across Italian municipalities demonstrated significant spatial spillovers and indirect effects that aspatial models entirely miss.

Limitations. Three limitations warrant acknowledgement. First, the Laplace approximation may lose accuracy when the conditional posterior of $\boldsymbol{\eta}$ is highly non-Gaussian (e.g. for very sparse data with many zero counts); higher-order Laplace corrections or adaptive quadrature could improve precision in such settings. Second, the unit-level random effects $\mathbf{u}$ are assumed independent Gaussian; correlated or non-Gaussian specifications would add flexibility at the cost of a more complex `TMB` template. Third, the eigenvalue pre-computation for the log-determinant $\log|\det(\mathbf{I} - \rho\mathbf{W})|$ is $O(n^3)$ and may become a bottleneck for very large n; stochastic log-determinant approximations [e.g. Barry and Pace, 1999] could be substituted.

Extensions. Several directions for future research emerge naturally. The most immediate is the incorporation of *temporal dependence*, yielding a spatio-temporal ZISAR of the form

$$\eta_{it} = \rho(\mathbf{W}\boldsymbol{\eta}_t)_i + \phi\,\eta_{i,t-1} + \mathbf{x}_{it}'\boldsymbol{\beta} + u_i + v_t,$$

where ϕ captures persistence and v_t is a time-specific effect. The panel structure introduces additional identification and computational challenges (dynamic spatial panel asymptotics, high-dimensional latent field) that merit a dedicated treatment.

A second extension concerns alternative spatial structures: replacing the single $\mathbf{W}$ with multiple weight matrices (e.g. geographic and economic proximity), or letting $\mathbf{W}$ be endogenous.

Third, the ZISAR template naturally accommodates other count distributions (generalised Poisson, Conway–Maxwell–Poisson) and alternative inflation links (probit, complementary log-log) by modifying the observation-model block of the `TMB` template.

Finally, packaging the estimator as a fully documented `R` package with formula interfaces, spatial-weights integration (`spdep`/`sf`), and automated model-selection tools would lower the barrier to adoption and is planned for a subsequent release.

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A Proofs

A.1 Proof of Proposition 1

We verify the conditions of the extremum-estimator consistency theorem [Newey and McFadden, 1994, Theorem 2.1].

Laplace approximation error. Let $\ell(\boldsymbol{\theta}) = \log \int p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\eta}$ denote the exact marginal log-likelihood and $\ell_{\mathrm{LA}}(\boldsymbol{\theta})$ the Laplace approximation (20). By Tierney and Kadane [1986], $|\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - \ell(\boldsymbol{\theta})| = O(1)$ uniformly over Θ (the error is bounded because the conditional density $f(y_i \mid \eta_i, \pi_i; \kappa)$ has bounded derivatives in η_i under Assumption 3). Dividing by n: $|n^{-1}\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - n^{-1}\ell(\boldsymbol{\theta})| = O(n^{-1}) \to 0$.

Uniform convergence. The exact marginal log-likelihood satisfies $n^{-1}\ell(\boldsymbol{\theta}) \to Q(\boldsymbol{\theta}) \equiv \mathbb{E}[\log p(y_1 \mid \boldsymbol{\theta})]$ uniformly over Θ, by the uniform law of large numbers. The key requirement is that the summands are bounded and equicontinuous in $\boldsymbol{\theta}$, which holds under the compactness of Θ (Assumption 3) and the boundedness of $\mathbf{W}$ (Assumption 1). Combined with the vanishing Laplace error, $\sup_{\boldsymbol{\theta}\in\Theta} |n^{-1}\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - Q(\boldsymbol{\theta})| \xrightarrow{p} 0$.

Identification. By Assumption 4, $Q(\boldsymbol{\theta})$ has a unique maximum at $\boldsymbol{\theta}_0$.

Conclusion. Compactness of Θ, uniform convergence, and identifiable uniqueness satisfy the conditions of the extremum-estimator theorem, giving $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$. □

A.2 Proof of Proposition 2

Score expansion. By a mean-value expansion around $\boldsymbol{\theta}_0$:

$$\mathbf{0} = n^{-1/2}\frac{\partial \ell_{\mathrm{LA}}}{\partial \boldsymbol{\theta}}\bigg|_{\hat{\boldsymbol{\theta}}} = n^{-1/2}\frac{\partial \ell_{\mathrm{LA}}}{\partial \boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}_0} + n^{-1}\frac{\partial^2 \ell_{\mathrm{LA}}}{\partial \boldsymbol{\theta}\,\partial \boldsymbol{\theta}'}\bigg|_{\bar{\boldsymbol{\theta}}} \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0).$$

CLT for the score. The Laplace-approximate score $\partial \ell_{\mathrm{LA}}/\partial \boldsymbol{\theta}$ can be decomposed as a sum of local contributions $s_i(\boldsymbol{\theta})$ that depend on unit i and its spatial neighbours. Under Assumption 1, the spatial weights have uniformly bounded row and column sums, so each s_i depends on at most $O(1)$ other units (the neighbourhood is bounded). This induces a mixing structure that satisfies the CLT for spatially dependent arrays [see Lee, 2004, Theorem 3.1]:

$$n^{-1/2}\sum_{i=1}^{n} s_i(\boldsymbol{\theta}_0) \xrightarrow{d} N\big(\mathbf{0},\, \boldsymbol{\Omega}_0\big),$$

where $\boldsymbol{\Omega}_0 = \lim_{n\to\infty} \mathrm{Var}\left[n^{-1/2}\sum_i s_i(\boldsymbol{\theta}_0)\right]$.

Hessian convergence. By the uniform LLN,

$$n^{-1}\frac{\partial^2 \ell_{\mathrm{LA}}}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}'}\bigg|_{\tilde{\boldsymbol{\theta}}} \xrightarrow{p} \mathcal{H}_0 \equiv \mathbb{E}\left[\frac{\partial^2 \log p(y_1 \mid \boldsymbol{\theta})}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}'}\bigg|_{\boldsymbol{\theta}_0}\right],$$

which is negative definite by Assumption 4. Under standard regularity (information matrix equality), $\mathcal{H}_0 = -\mathcal{I}(\boldsymbol{\theta}_0) = -\boldsymbol{\Omega}_0$.

Combining. Rearranging the score expansion:

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\mathcal{H}_0^{-1}\, n^{-1/2}\sum_i s_i(\boldsymbol{\theta}_0) + o_p(1) \xrightarrow{d} N\big(\mathbf{0},\, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}\big).$$

□

A.3 Proof of Proposition 3

Under the restricted ZISAR ($\mathbf{g} = \mathbf{0}$), the marginal expectation of y_i evaluated at the conditional mode is

$$\mathbb{E}[y_i] \approx (1-\pi_i)\,\mu_i, \qquad \mu_i = \exp(\eta_i), \quad \pi_i = \mathrm{logit}^{-1}(\mathbf{z}_i'\boldsymbol{\gamma}).$$

The latent predictor at the mode satisfies $\hat{\boldsymbol{\eta}} \approx \mathbf{S}(\rho)\mathbf{X}\boldsymbol{\beta}$ (ignoring the random-effect contribution, which integrates out at the mode). The partial derivative of $\boldsymbol{\eta}$ with respect to x_{jr} is

$$\frac{\partial \eta_i}{\partial x_{jr}} = [\mathbf{S}(\rho)]_{ij}\,\beta_r.$$

The effect on $\mathbb{E}[y_i]$ is

$$\frac{\partial \mathbb{E}[y_i]}{\partial x_{jr}} = (1-\pi_i)\,\mu_i\,[\mathbf{S}(\rho)]_{ij}\,\beta_r + \underbrace{(\text{inflation channel})}_{\text{diagonal only}}.$$

Stacking: $\mathbf{M}_r = \mathrm{diag}((1-\hat{\boldsymbol{\pi}}) \odot \hat{\boldsymbol{\mu}})\,\mathbf{S}(\rho)\,\beta_r$.

(i) When $\rho \in [0,1)$ and $\mathbf{W} \geq 0$ entrywise, the power-series $\mathbf{S}(\rho) = \sum_{s=0}^{\infty}\rho^s\mathbf{W}^s$ has all non-negative entries. Since $(1-\pi_i)\mu_i > 0$, $\mathbf{M}_r$ has the same sign as β_r.

(ii) When $\rho = 0$, $\mathbf{S}(\rho) = \mathbf{I}_n$ and $\mathbf{M}_r$ is diagonal, so all indirect effects (off-diagonal elements) are zero. For $\rho \neq 0$, $\mathbf{S}(\rho)$ has positive off-diagonal elements (since $\mathbf{W}^s$ has positive entries for $s \geq 1$ on connected lattices), making indirect effects non-zero.

(iii) Each element of $\mathbf{S}(\rho) = \sum_{s=0}^{\infty} \rho^s \mathbf{W}^s$ is monotonically increasing in $|\rho|$ for $\rho \geq 0$ (since all terms $\rho^s [\mathbf{W}^s]_{ij}$ are non-negative and increasing in ρ). □

B Sensitivity Analysis

Table 8 reports ZISAR estimates under alternative spatial weights specifications for the empirical application (Section 6).

Table 8: Sensitivity of ZISAR estimates to the choice of spatial weights.

	k-NN ($k = 4$)	k-NN ($k = 6$)	k-NN ($k = 8$)
$\hat{\rho}$	*to be filled from R output*		
$\hat{\beta}_1$ (pop)			
$\hat{\beta}_2$ (aging)			
$\hat{\kappa}$			
AIC			

C TMB Implementation Details

The `TMB` template declares $\boldsymbol{\eta}$ as random effects and uses the SAR prior density:

$$-\log p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma) = \frac{n}{2}\log(2\pi\sigma^2) - \log|\det(\mathbf{I} - \rho\mathbf{W})| + \frac{1}{2\sigma^2}\|(\mathbf{I} - \rho\mathbf{W})\boldsymbol{\eta} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

The log-determinant is computed via pre-computed eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathbf{W}$ (passed as data). For possibly complex eigenvalues ($\lambda_j = a_j + b_j \mathrm{i}$):

$$\log|\det(\mathbf{I} - \rho\mathbf{W})| = \frac{1}{2}\sum_{j=1}^{n} \log\left[(1 - \rho a_j)^2 + (\rho b_j)^2\right].$$

The sparse matrix–vector product $(\mathbf{I} - \rho\mathbf{W})\boldsymbol{\eta}$ is computed via Eigen's sparse algebra, preserving the $O(\mathrm{nnz}(\mathbf{W}))$ complexity. The reparametrisation $\rho = \tanh(\tilde{\rho})$ ensures $\rho \in (-1, 1)$ without box constraints. Standard errors on the original scale are obtained by the delta method through `TMB`'s `ADREPORT` mechanism.